



UNIVERSITY OF
COPENHAGEN

BAYESIAN OPTIMIZATION OF GPU-ACCELERATED OCEAN MODELS

Marta Agnieszka Mrozowska¹, Aster Stoustrup¹, James Avery^{1,2}, Roman Nuterman¹, Carl-Johannes Johnsen² and Markus Jochum¹

¹Niels Bohr Institute, University of Copenhagen, Denmark; ²Department of Computer Science, University of Copenhagen, Denmark

BAYESIAN OPTIMIZATION IN A NUTSHELL

BO solves the problem:

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} f(\theta)$$

by constructing a **probabilistic surrogate model** of $f(\theta)$. The model is used to compute an **acquisition function**, which is then minimized. The acquisition function **minima are evaluated** and used to update the **surrogate model posterior**. This makes the method **highly sample-efficient**.

INTRODUCTION

As climate models become more complex, the need for **automated, reproducible** methods for **parameter calibration** grows. The **Versatile Optimizer (VerOpt)** is a Python package adapting Bayesian Optimization (BO) to climate model calibration. The aim of VerOpt is to be **user- and BO beginner-friendly**, and to offer **high flexibility** in optimization design.

This poster presents the VerOpt workflow with an example where we aim to reduce annual mixed layer depth (MLD) biases in the JAX-based ocean GCM Veros. The optimizer finds a combination of parameter values that reduce the MLD biases by 23%.

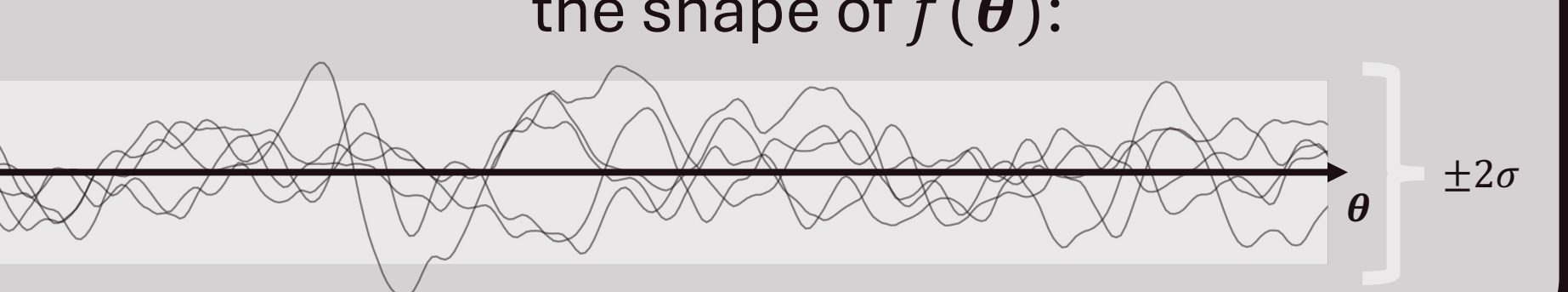
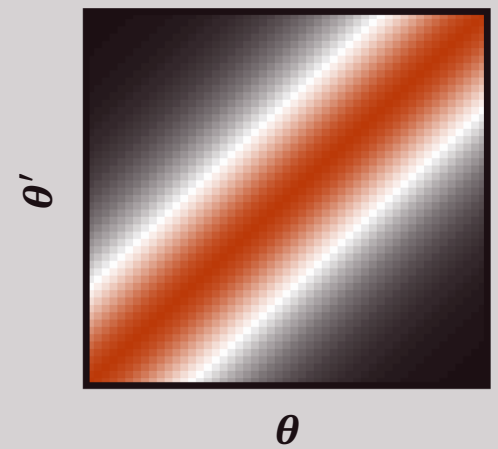
PROBLEM

The **9-dimensional parameter space** consists of vertical mixing scheme, mesoscale eddy closure and surface forcing parameters.

Veros 1° simulations run for **60 years** and the last decade is used for analysis. The target MLD climatology is from de Boyer Montégut (2022).

SETUP

The Gaussian surrogate model is initialized with the **Matérn kernel**, $\mu = \mathbf{0}$ and $\sigma = 1$. Random draws from the model illustrate our **prior** guess about the shape of $f(\theta)$:

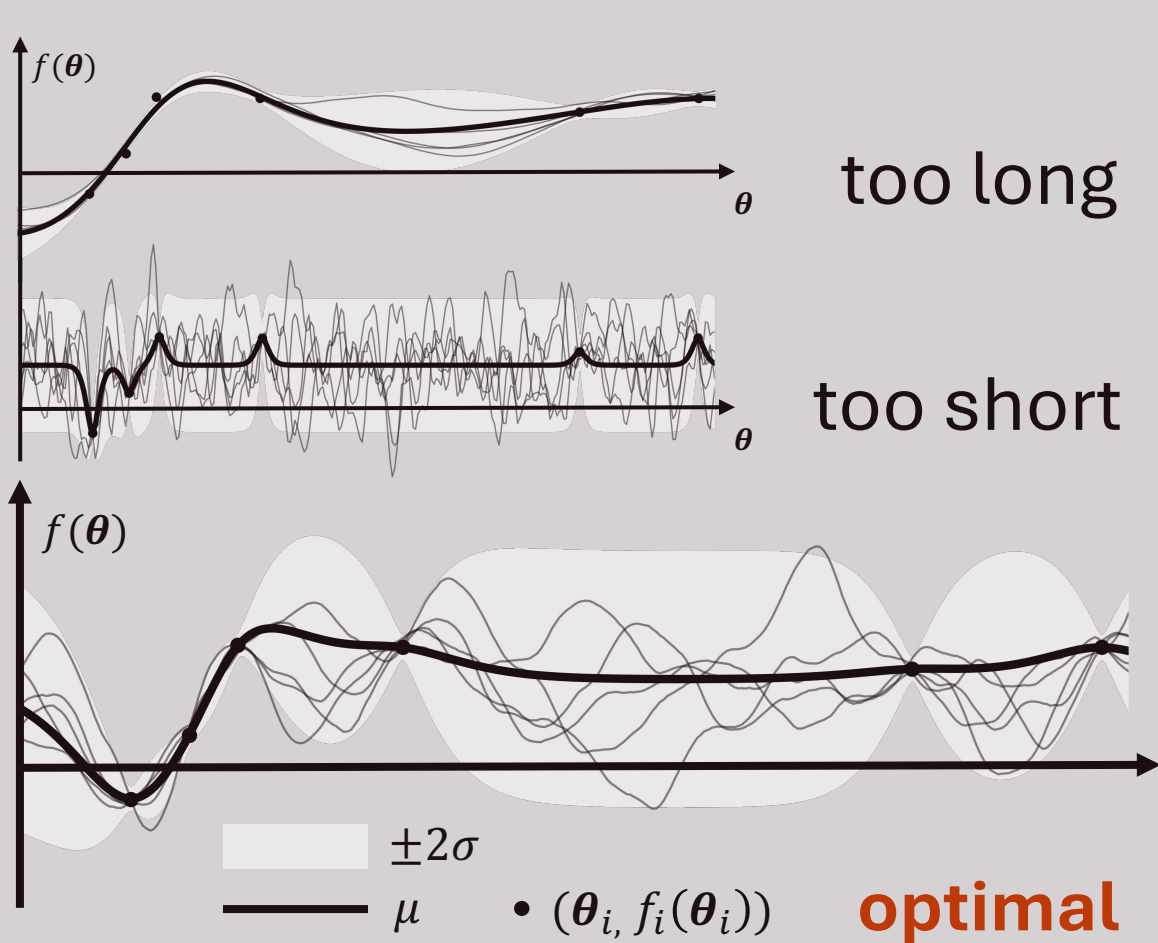


INITIALIZE OPTIMIZATION

At each optimization round, the optimizer suggests a batch of $N = 32$ parameter vectors. The parameter values in the initial batch are **randomly selected**.

SURROGATE MODEL

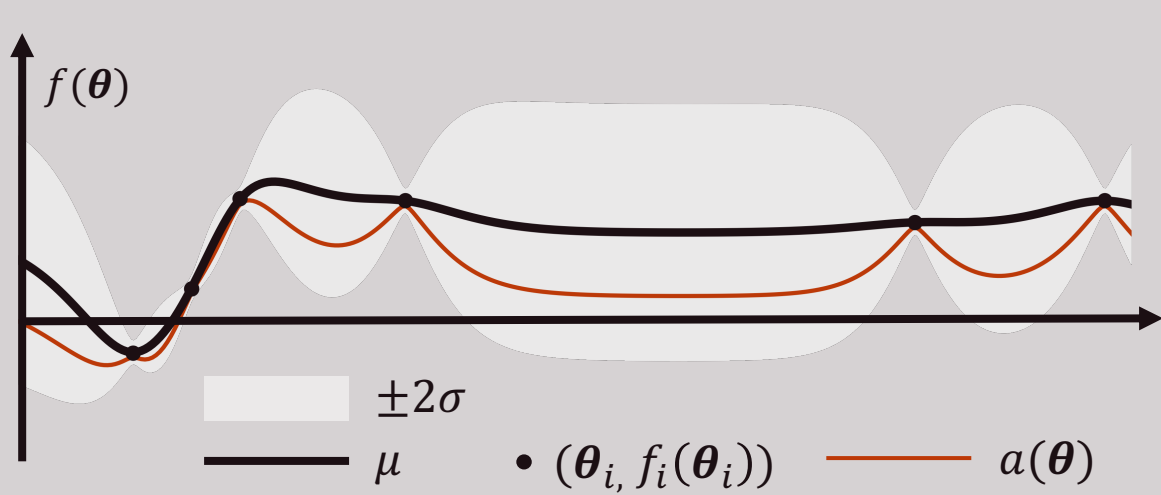
The model training determines the **length scale** that best fits the evaluated points.



ACQUISITION FUNCTION

The optimal model is used to construct the acquisition function. Here, we use **lower confidence bound (LCB)**:

$$a(\theta) = \mu - \beta\sigma.$$

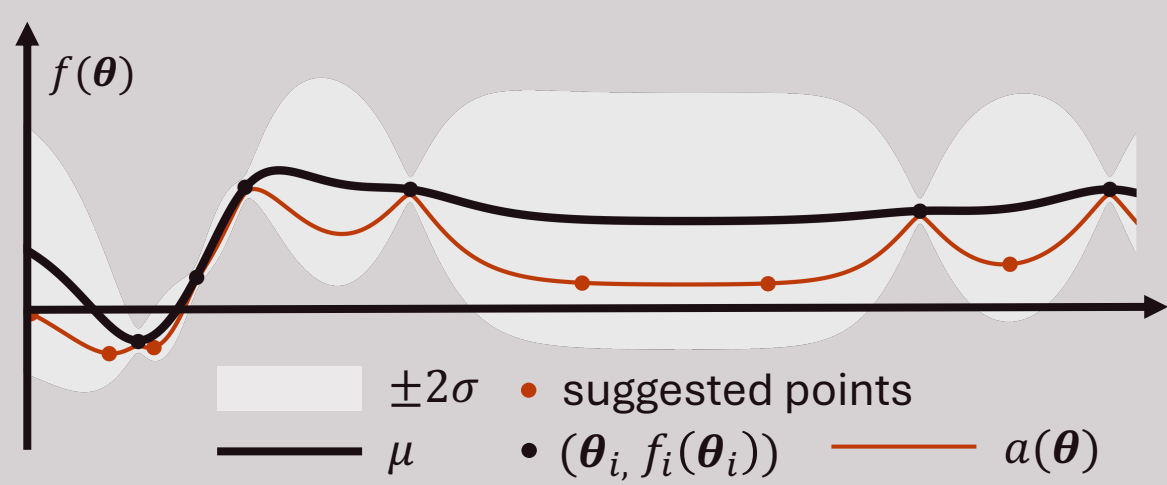


Higher β will force the optimizer to search in uncertain regions of the parameter space.

SUGGESTED POINTS

The acquisition function is **minimized**. The minima form the new set of N parameter vectors.

The optimization loop restarts with new parameters.



VEROPT OPTIMIZATION LOOP

New objective function values

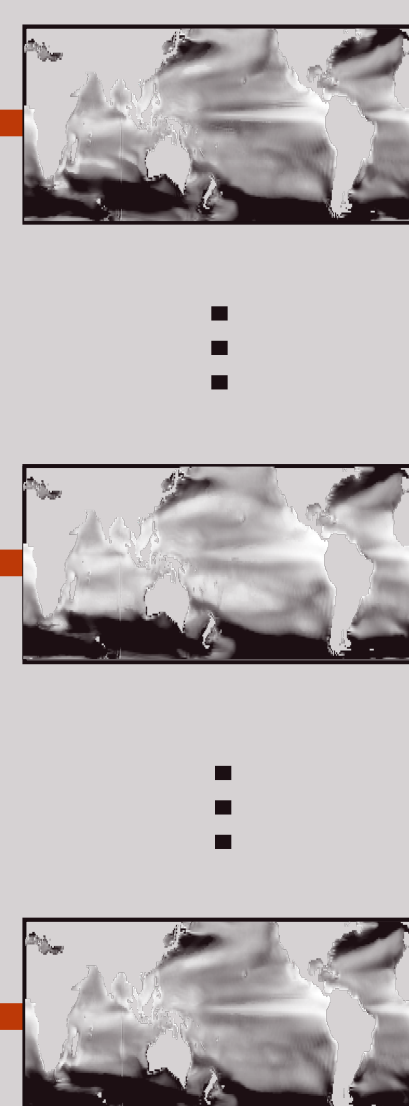


PROCESSING

$$f_i(\theta_i) = \operatorname{rms}\left(\frac{y_i - y_{obs}}{y_{obs}}\right) + \dots$$

y_i : simulated MLD
 y_{obs} : MLD climatology

The full objective also includes thermocline depth and circulation strength biases.

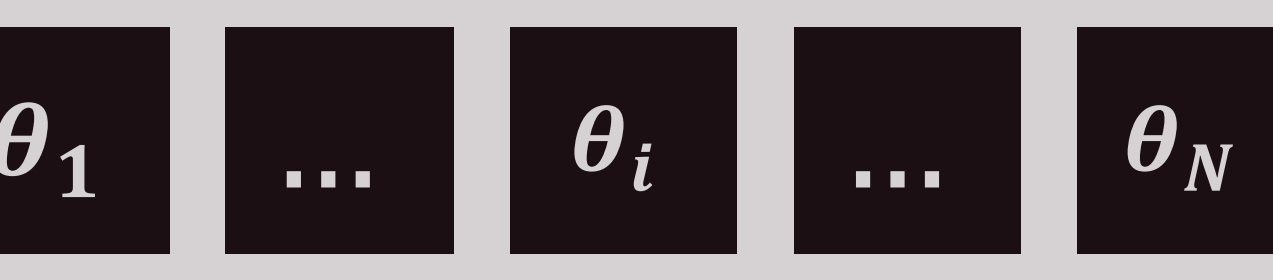


SUBMISSION

The 32 Veros simulations run in parallel on the LUMI HPC GPU nodes.

In comparison to performance on CPU, the GPU simulations **use 17 times less energy** and **reduce simulation time by an order of magnitude**.

New parameters to evaluate



VEROPT INTERFACES

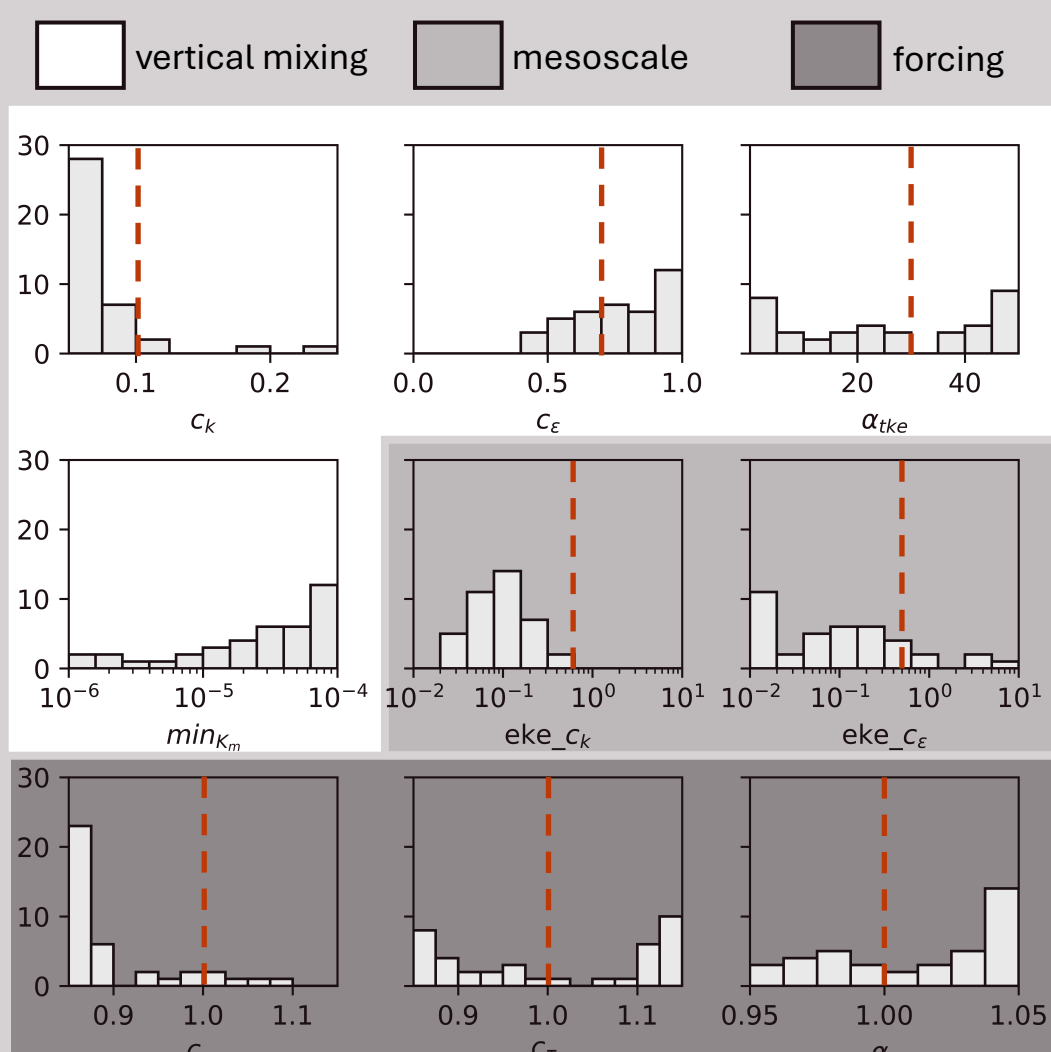
automatically submit simulations either **locally** or on **slurm**. The simulations are **tracked** and **processed** when they finish, accounting for diverged solutions.

RESULTS

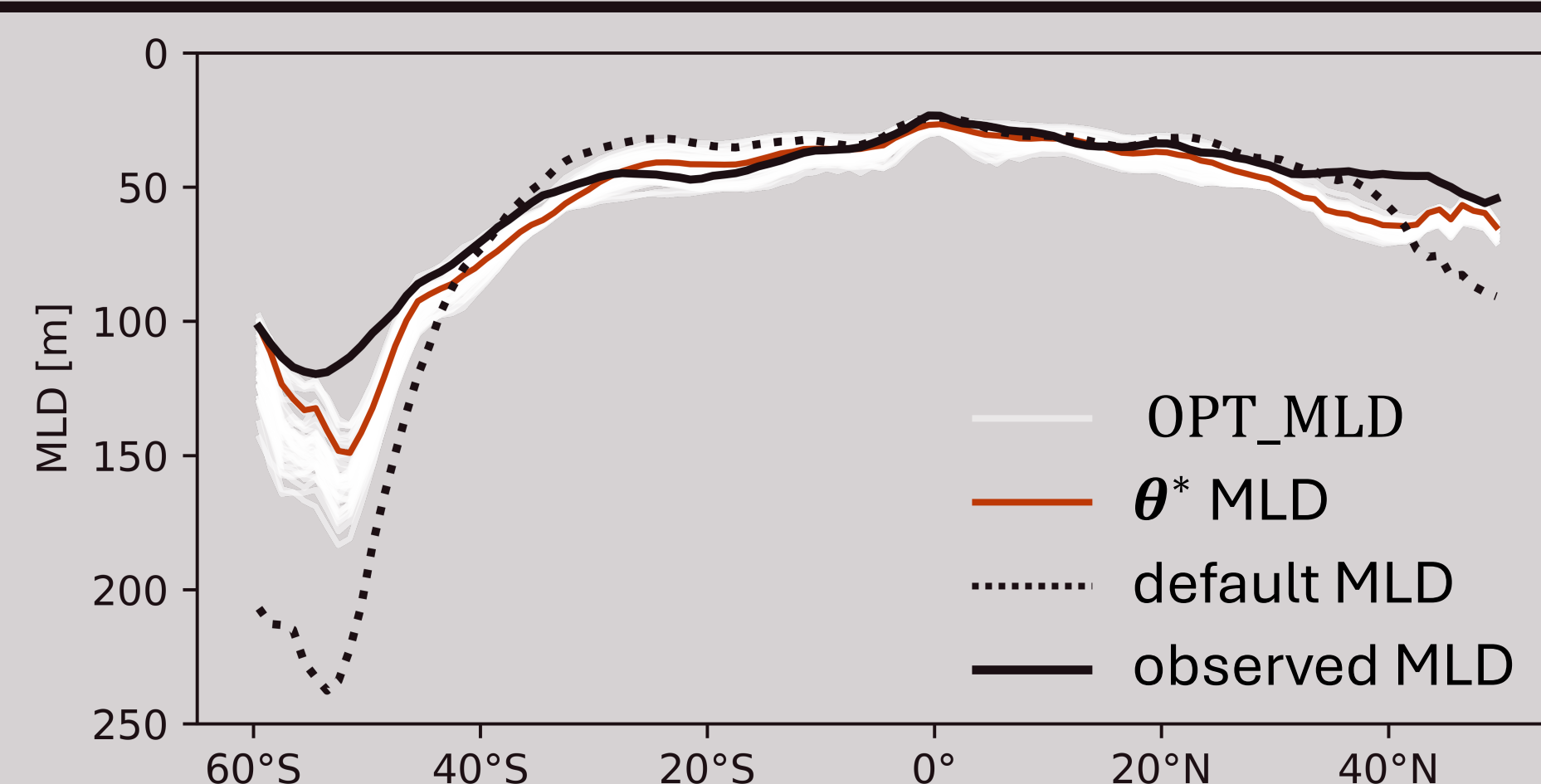
After **17 optimization rounds** of and **375 completed Veros runs**, the best set of parameters θ^* reduces Veros MLD biases by 23%.

The optimal solutions for which $\operatorname{rmse}(\text{MLD})$ is smaller than 45% (OPT_MLD) share the following properties:

- Zonal wind stress is 10% weaker**
- GM coefficient is halved**

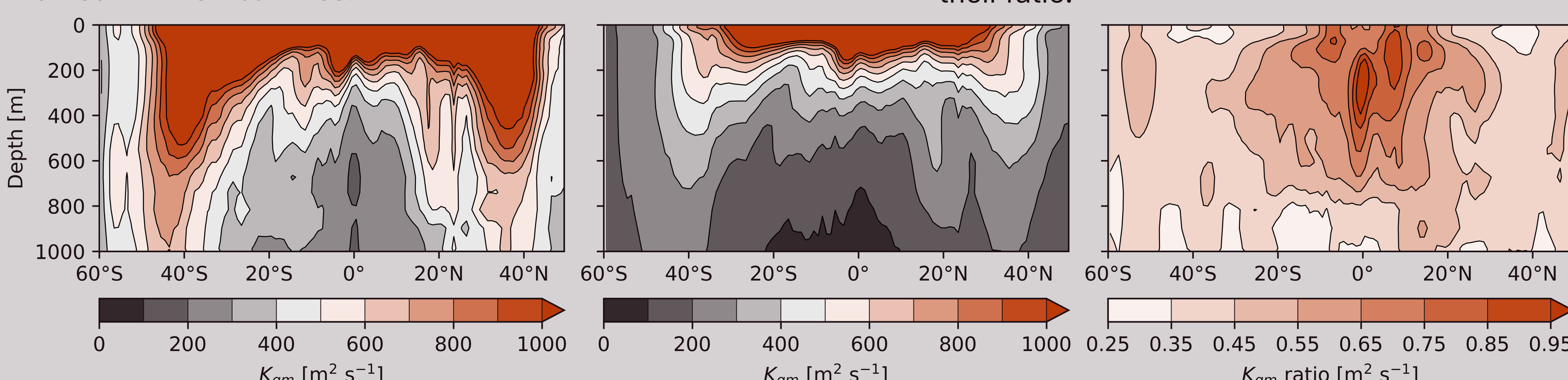


Top: OPT_MLD parameter value distributions. The default values are marked with vertical lines.



Top: zonal average MLD in the OPT_MLD simulations;

Bottom left: GM coefficient in the default simulation; **middle:** OPT_MLD average and **right:** their ratio.



UPCOMING

Community ESM (Shields et al. 2012)
Tunax mixed layer model
(Mouttapa et al. 2025)
Parallel Ice Sheet Model
(Khrulev et al. 2023)

**VEROPT 1.0 RELEASE
ON JULY 10TH!**



DISCLAIMER: This poster is fully human-made. The generator was my brain interfacing with PowerPoint enhanced by two rounds of prompts from the co-authors. The generation took four days and the NN training spanned over a decade.

REFERENCES:

de Boyer Montégut, C. (2022) *Mixed layer depth climatology computed with a density threshold criterion of 0.03kg/m3 from 10 m depth value*. SEANOE. [10.17882/91774](https://doi.org/10.17882/91774)
Khrulev, C. et al. (2023) *Parallel Ice Sheet Model (PISM) (v2.1)*. Zenodo. [10.5281/zenodo.10202029](https://doi.org/10.5281/zenodo.10202029)
Mouttapa, G. et al. (2025) *Leveraging automatic differentiation for calibrating vertical mixing parameterizations*. EGU General Assembly 2025. [10.5194/egusphere-egu25-17338](https://doi.org/10.5194/egusphere-egu25-17338)
Shields, C. et al. (2012) *The Low-Resolution CCSM4*. Journal of Climate. [10.1175/JCLI-D-11-00260.1](https://doi.org/10.1175/JCLI-D-11-00260.1)